

CUSTOMER SEGMENTATION PROJECT

Using K-Means Clustering & RFM Analysis

Submitted by

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1. Project Overview

This project focuses on Customer Segmentation for an Online Retail Store using the K-Means Clustering algorithm combined with RFM (Recency, Frequency, Monetary) Analysis. The goal is to group customers into meaningful segments based on their purchase behavior — without any predefined labels — so that the business can design targeted marketing campaigns for each group.

The dataset used is the Online Retail dataset containing approximately 541,909 transaction records from a UK-based online retailer between December 2010 and December 2011. The analysis was performed entirely using Python in Jupyter Notebook with libraries including pandas, numpy, sklearn, matplotlib, and seaborn.

2. Dataset Overview

The dataset contains the following 8 columns:

Column Name	Description
InvoiceNo	Unique invoice number of each transaction
StockCode	Unique product code
Description	Product name
Quantity	Number of items purchased per transaction
InvoiceDate	Date and time of the transaction
UnitPrice	Price per unit in British Pounds (£)
CustomerID	Unique identifier for each customer
Country	Country where the customer resides

3. Data Exploration & Cleaning

Before building the model, the raw data was explored and cleaned through the following steps:

- Total raw rows: 541,909 | Total columns: 8
- Missing values found: CustomerID had 135,080 nulls; Description had 1,454 nulls
- Negative Quantity values (returns/cancellations) were removed
- Zero or negative UnitPrice rows were removed
- Cancelled invoices (InvoiceNo starting with 'C') were removed
- Final clean dataset: 397,884 rows after all cleaning steps

4. RFM Feature Engineering

Since the dataset contains transaction-level data (one row per product per invoice), three meaningful customer-level features were engineered using the RFM methodology:

Feature	Definition	How Calculated
Recency (R)	How recently the customer made a purchase	Days between last purchase date and reference date (2011-12-10)
Frequency (F)	How many times the customer purchased	Count of unique InvoiceNo per CustomerID
Monetary (M)	How much total money the customer spent	Sum of (Quantity x UnitPrice) per CustomerID

After aggregation, the RFM table contained 4,338 unique customers. Outliers were removed using the IQR method ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$), reducing the dataset to 3,710 customers. Features were then scaled using StandardScaler to ensure equal contribution to the distance-based K-Means algorithm.

5. Project Questions & Answers

5.1 Data Exploration

Q: How many total customers are present in the dataset?

A: After removing rows with missing CustomerID values, there are 4,338 unique customers in the cleaned dataset. After applying the IQR outlier removal method, 3,710 customers were used for the final clustering model.

Q: How many features are used for clustering after preprocessing?

A: 3 features were used for clustering — Recency, Frequency, and Monetary (RFM). These were derived from the original 8-column raw transaction data through feature engineering and aggregation at the CustomerID level.

Q: Which feature has the highest variance among all features?

A: Monetary has the highest variance among the three RFM features. This is because customer spending varies drastically — from as low as £3.75 to over £3,600 after outlier removal — reflecting the wide diversity in purchasing power and behavior across customers.

5.2 Cluster Formation

Q: How many clusters were finally created by the clustering algorithm?

A: 3 clusters were created. The optimal $K=3$ was determined using the Elbow Method (sharp bend observed at $K=3$) and validated by the Silhouette Score which peaked at $K=3$ with a score of 0.4600.

Q: What is the size (number of customers) of each cluster?

A: Regular Customers (Cluster 0): 1,968 customers | At-Risk Customers (Cluster 1): 920 customers | Premium Customers (Cluster 2): 822 customers. Total: 3,710 customers.

Q: Which cluster contains the highest number of customers?

A: Cluster 0 — Regular Customers — contains the highest number of customers with 1,968 customers, representing approximately 53% of the total customer base.

Cluster	Label	No. of Customers	% of Total
Cluster 0	Regular Customers	1,968	53%
Cluster 1	At-Risk Customers	920	25%
Cluster 2	Premium Customers	822	22%

5.3 Cluster Characteristics

Q: What is the average RFM of customers in each cluster?

A: The cluster-wise average RFM values are shown in the table below:

Cluster	Label	Avg Recency (days)	Avg Frequency	Avg Monetary (£)
Cluster 0	Regular Customers	48.78	2.12	598.79
Cluster 1	At-Risk Customers	226.80	1.50	414.09
Cluster 2	Premium Customers	36.23	6.05	2,107.88

Q: Which cluster has the highest average spending score?

A: Cluster 2 — Premium Customers — has the highest average spending with £2,107.88. These customers also have the lowest recency (36 days) and highest purchase frequency (6.05 times), making them the most valuable segment.

Q: Which cluster represents the lowest spending customers?

A: Cluster 1 — At-Risk Customers — represents the lowest spending group with an average monetary value of £414.09. They also have the highest recency (226.80 days), meaning they haven't purchased in over 7 months, and the lowest frequency (1.50 times).

5.4 Model Evaluation

Q: What is the inertia value obtained from the K-Means clustering model?

A: The inertia value for the final K-Means model with K=3 is 3,502.54. Inertia measures the sum of squared distances of each data point from its assigned cluster centroid — a lower value indicates tighter, more compact clusters.

Q: What is the silhouette score of the clustering result?

A: The silhouette score is 0.4600. This score ranges from -1 to +1, where values closer to 1 indicate well-separated and cohesive clusters. A score of 0.46 is considered good for real-world customer data, confirming that the 3 clusters are meaningfully distinct.

Q: Based on the elbow method, what is the optimal number of clusters?

A: The optimal number of clusters is K=3. The elbow curve showed a sharp drop in inertia from K=1 (11,130) to K=3 (3,502), after which the rate of decrease slowed significantly. The silhouette score also confirmed K=3 as optimal with the highest score of 0.4600.

K Value	Inertia	Silhouette Score
K=1	11,130.00	—
K=2	6,015.49	0.4541

K Value	Inertia	Silhouette Score
K=3	3,502.54	0.4600 ← Optimal
K=4	2,706.40	0.4133
K=5	2,321.45	0.4004

5.5 Visualization-Based Questions

Q: In the cluster visualization plot, which cluster appears most densely populated?

A: In both the Recency vs Monetary and Frequency vs Monetary scatter plots, Cluster 0 (Regular Customers, shown in green) appears as the most densely populated cluster. This is consistent with the actual count — 1,968 customers representing 53% of the total segmented base.

Q: Which cluster represents customers with high income but low spending score?

A: In the Recency vs Monetary scatter plot, the At-Risk Customers (Cluster 1, red dots) are positioned on the far right with low monetary values. These customers made purchases in the past but have not returned for a long time (average 226 days) and spent the least on average (£414.09), representing the low-engagement, low-spending segment.

Q: Which cluster represents low recency but high spending customers?

A: Premium Customers (Cluster 2, blue dots) represent low recency (bought very recently — average 36 days ago) combined with the highest monetary value (£2,107.88 average). In the Recency vs Monetary chart, they are visibly concentrated in the top-left zone — low recency on X-axis and high monetary on Y-axis.

5.6 Cluster Label Analysis

Q: Which cluster would most likely represent premium customers?

A: Cluster 2 represents Premium Customers. This cluster has the lowest Recency (36 days — bought most recently), the highest Frequency (6.05 purchases), and the highest Monetary value (£2,107.88). These customers are actively engaged, loyal, and spend significantly more than others. From a business perspective, they deserve VIP treatment, loyalty rewards, and early access to new products.

Q: Which cluster might correspond to price-sensitive customers?

A: Cluster 1 (At-Risk Customers) corresponds to price-sensitive customers. They have the highest recency (226 days — last purchased almost 7.5 months ago), lowest frequency (1.50), and lowest spending (£414.09). These customers may have been deterred by pricing, found alternatives, or simply disengaged. They require win-back campaigns with discounts or special offers to re-engage.

5.7 Outlier Detection

Q: Are there any potential outliers visible in the dataset after clustering?

A: Yes, potential outliers are visible after clustering. Before outlier removal using the IQR method, the raw RFM table showed extreme values — Monetary reached up to £280,206, Frequency up to 209 purchases, and Recency up to 374 days. The IQR method removed 628 such extreme customers (from 4,338 to 3,710) before clustering. After clustering, a few boundary customers with unusually high distances from their cluster centroids are visible as outlier markers in the scatter plot.

Q: Which data points appear farthest from cluster centroids?

A: After calculating Euclidean distances from each customer to their assigned cluster centroid using the `cdist` function from `scipy`, the top 10 customers with the highest distance values were identified as the farthest points. These customers typically showed extreme values in Monetary or Recency — spending far more than their cluster average or having very unusual purchase timing. In the outlier detection scatter plot, these customers are highlighted as black X markers sitting at the edges or boundaries between clusters, visually confirming their outlier nature.

5.8 Centroid Analysis

Q: What are the centroid values for each cluster?

A: The centroid values (in scaled form after `StandardScaler`) for each cluster are shown below:

Cluster	Label	Recency (Scaled)	Frequency (Scaled)	Monetary (Scaled)
Cluster 0	Regular Customers	-0.4650	-0.3262	-0.3542
Cluster 1	At-Risk Customers	+1.5372	-0.6098	-0.5818
Cluster 2	Premium Customers	-0.6075	+1.4657	+1.5015

Note: Scaled values are in standard deviation units. Positive values mean above average; negative values mean below average for that feature.

Q: Which cluster centroid has the highest average monetary value?

A: Cluster 2 (Premium Customers) has the highest monetary centroid value of +1.5015 in scaled form, corresponding to an actual average monetary value of £2,107.88. This is significantly higher than Cluster 0 (£598.79) and Cluster 1 (£414.09), confirming Cluster 2 as the most financially valuable customer segment.

6. Conclusion & Business Recommendations

The K-Means clustering model successfully segmented 3,710 customers into 3 distinct groups based on their RFM behavior. The silhouette score of 0.4600 confirms that the clusters are well-separated and meaningful for business decision-making.

Segment	Size	Key Characteristics	Recommended Action
Premium Customers	822 (22%)	Recent buyers, high frequency, highest spend	VIP rewards, early access, loyalty programs
Regular Customers	1,968 (53%)	Moderate recency, low-mid frequency, moderate spend	Upsell campaigns, engagement offers, product recommendations
At-Risk Customers	920 (25%)	Inactive for 7+ months, lowest frequency and spend	Win-back emails, discount offers, re-engagement campaigns

This segmentation enables the marketing team to allocate budget efficiently — investing more in retaining Premium customers, converting Regular customers to Premium, and re-activating At-Risk customers before they are permanently lost.

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